

WELCOME TO CSE 231: Circuits & Electronics

Course Syllabus

Class Time: **Tuesday 10:30 – 11:30**
 Thursday 12:30 - 15:30

Instructor: **Erkan Zergeroğlu, PhD.**
 Office: 202 Computer Engineering Department
 E-mail: e.zerger@gtu.edu.tr

Office Hours: **Yet to be determined**

Textbook

- I. The essentials of electric circuits / M. Fogiel. Fogiel**
- II. Electric circuits fundamentals / Thomas L. Floyd. 1998**
- III. Introduction to electric circuits / Richard C. Dorf. 2001**
- IV. Principles of electric circuits / Thomas L. Floyd. 2000**
- V. Electric circuits / James W. Nilsson, Susan A. Riedel. 2008**
- VI. Fundamentals of electric circuits / Charles K. Alexander, Matthew N.O. Sadiku. 2000 I**

Homework

- Homework will be assigned and collected.
- Working the homework problems is essential to the learning of the material in this course; in fact, most of your learning will come from doing the homework.
- It is expected that your homework will represent your own work, although working in groups is allowed, and even encouraged.
- Late homework will not be accepted

Exams

There will be at least one regular (midterm) closed book exam and a “face to face, on campus ” final exam.

- Most questions will be circuit analysis problems, including numerical as well as symbolic answers; however, there may be a few conceptual questions as well on each exam.**

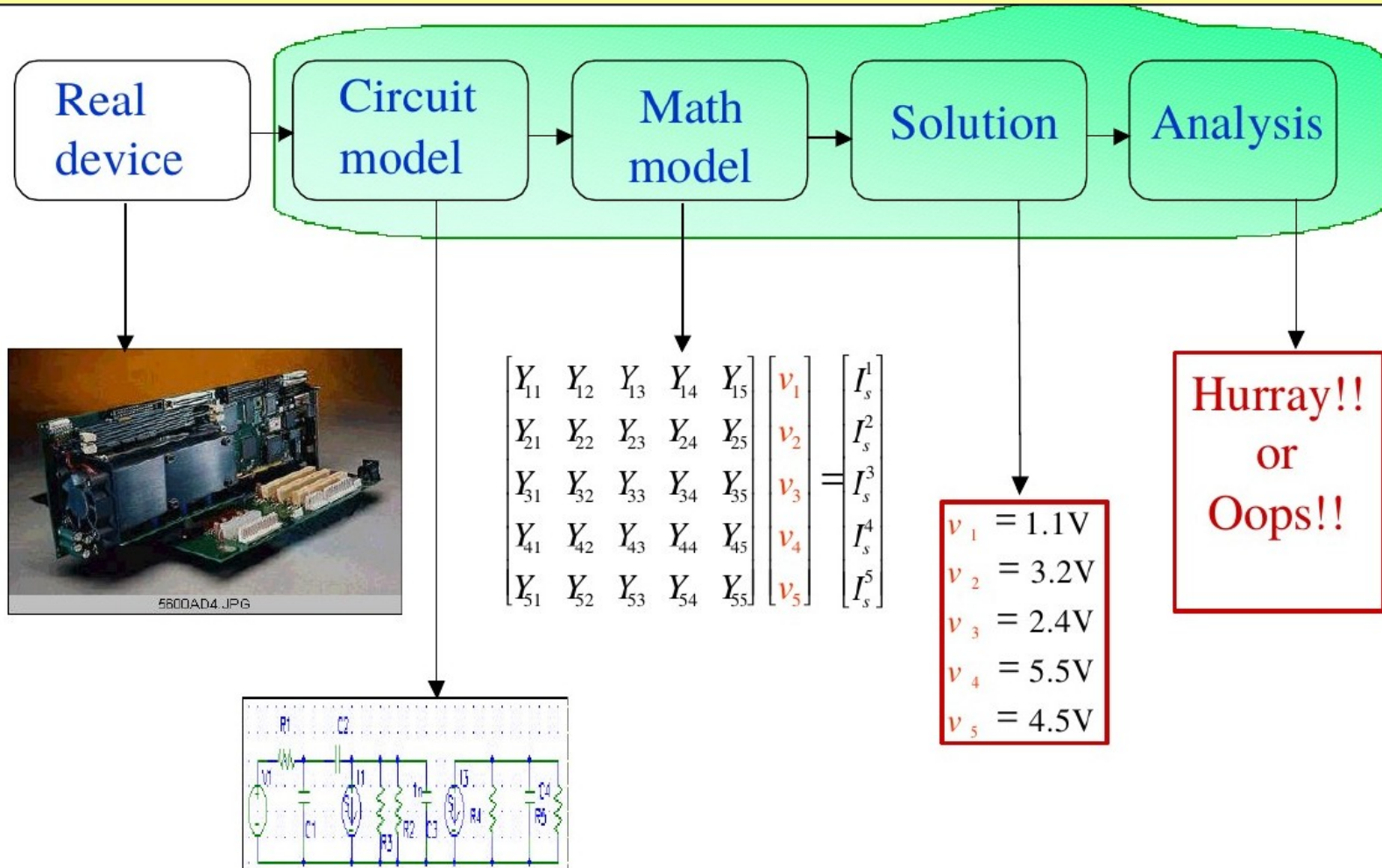
Grading

**Final Grades will be Determined by Averaging the Homework, Exams, and the Final Exam Based on the Following Scale
(This part is subject to change):**

Quizzes	10%
Homework	10%
Midterm Exam	30 %
Final Exam	50% (min 30 out of 100 is required)

Course Grade	100%

Overall



Lecture 1

Circuit Variables

Motivation

- Basis for future courses
- Foundational to Electrical Engineering and Computer Engineering
- Used for actual circuits and circuit models

Circuit Theory

Assumptions:

- Electrical effects are instantaneous
- No magnetic coupling between components



Lumped-Parameter Model

Circuit Variables

Units and Dimensions

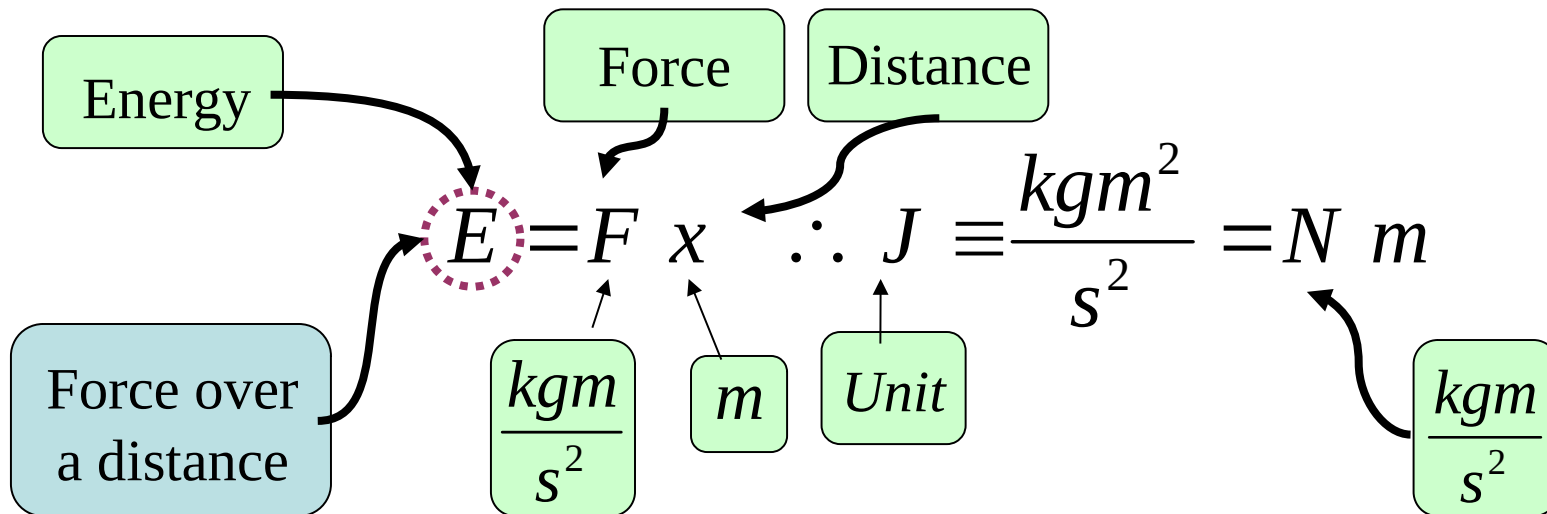
We will use SI Units [International System]

- Length m
- Mass kg
- Time s
- Current Ampere
- Temperature K
- Voltage Volts
- Resistance Ohms

Some Important Derived Units

Energy

Units are *Joules* (J)



Note: Text uses “ W ” for Energy

Some Important Derived Units (Contd.)

Power

Units are **Watts** (W)

$$W = \frac{J}{s} \equiv \frac{\text{Energy}}{\text{Time}} = \frac{W}{t} \Rightarrow \frac{dW}{dt} = \frac{\text{Change in } W}{\text{Change in } t}$$

$$J \equiv W \cdot s \quad \text{OR} \quad kW\text{-hr}$$

Meters read energy
Power companies charge for energy use

Some Important Derived Units (Contd.)

Charge

Units are **Coulombs** (C)

$$\text{Coulomb} \equiv A \cdot s \equiv \text{Current} \times \text{Time} \} \text{ Charge}$$

OR

$$\text{Current} \equiv \left(\frac{C}{s} \right) \equiv \frac{\text{Charge}}{\text{Time}}$$

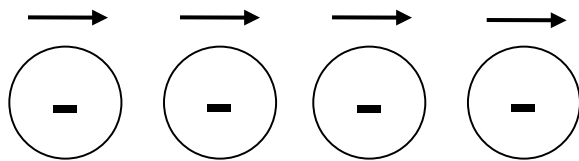
Related to velocity of electrons

Definition of Current

Current is charge in motion

$$i = \frac{dq}{dt} \quad (\text{Coulombs/second}) \equiv \text{Amperes} = \frac{\text{Change in } q}{\text{Change in } t}$$

$$q = \int_0^t i \, dt \quad \therefore Q_{\text{total}} = \int_0^{\infty} i \, dt \quad \left. \vphantom{\int_0^{\infty} i \, dt} \right\} \begin{array}{l} \text{Integrate above} \\ \text{expression} \end{array}$$



Current in Amperes or Amps
 $\equiv$ # Coulombs which cross
a given point in 1 second

Electrons moving or flowing through a wire is a current
These electrons are “moved” by a voltage

Some Important Derived Units (Contd.)

Voltage

Definition of Voltage

$$V \equiv \frac{w}{q} \equiv \frac{\text{Joules}}{\text{Coulomb}} \\ \equiv \text{Voltage}$$

$V_{a-b} \equiv$ Work required to move charge from point “a” to “b”
(Book uses energy)

$$V = \frac{dw}{dq} = \frac{\text{Change in } w}{\text{Change in } q}$$

Some Important Derived Units (Contd.)

Resistance

Units are *Ohms* (Ω)

$$\Omega = \frac{V}{A} \quad \text{Ohm's Law}$$

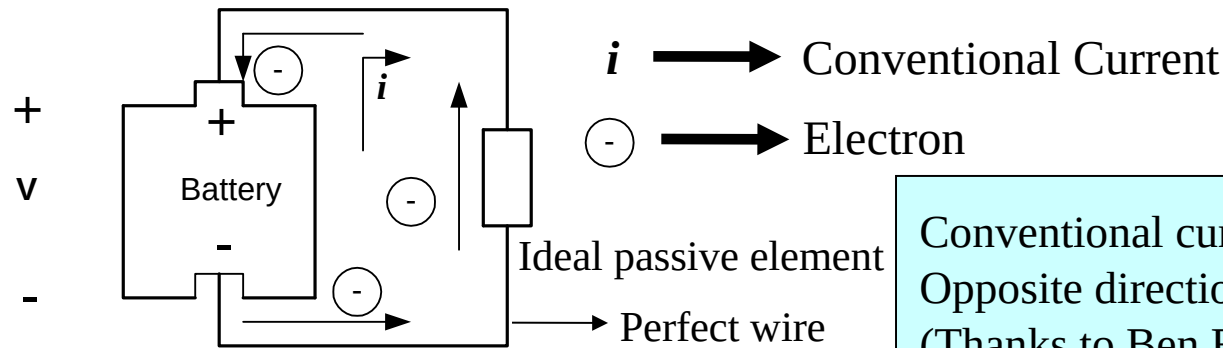
$$\Rightarrow \text{Implication: } V = IR$$

$$\text{Voltage} = \text{Current} \times \text{Resistance} \quad \left. \vphantom{\text{Voltage} = \text{Current} \times \text{Resistance}} \right\} \text{Ohm's Law}$$

Voltage and Current

Separation of Charge $\longrightarrow$ Force between charge $\longrightarrow$ **Voltage**

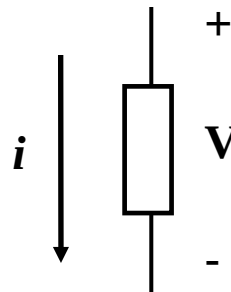
Voltage moves electrons $\longrightarrow$ **Current**



Conventional current, “ i ”, flows in the Opposite direction to the electrons (Thanks to Ben Franklin)

- + Terminal attracts electrons
- Terminal repels electrons

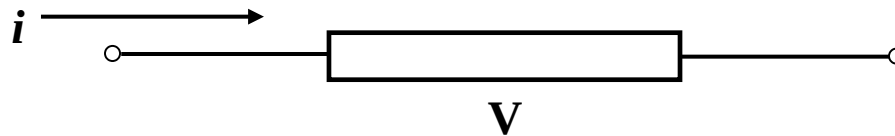
Ideal Passive Element
(seen above)



i flows “**through**” the element

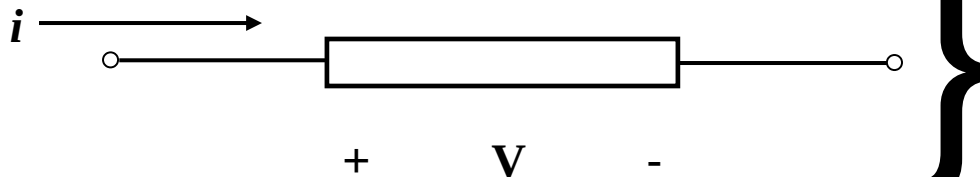
V is “**across**” the element
(drop or rise)

Passive Sign Convention



What are the polarities?

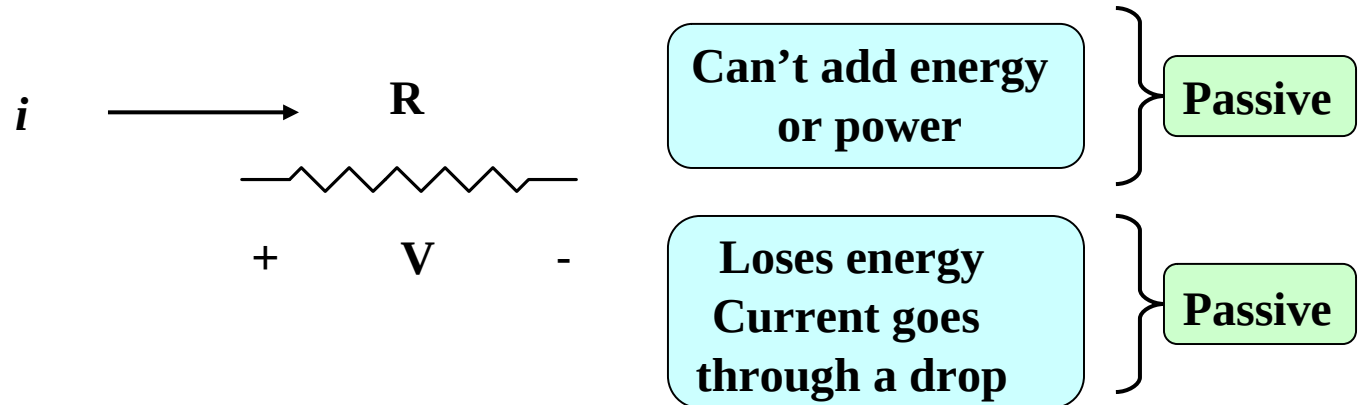
- The current i flows to the right
- What is the polarity of V ?
- General element, so we don't know really



Passive sign convention

Variables are “Positive” in the equations

Passive Sign Components



This is not reality for “Active” Components

- Does add energy or supply power
 - Current goes through a rise $\Rightarrow$ Gain in energy
- The two bullet points are grouped by a right-facing curly brace, which points to a green box labeled "A battery is an active component".

Convention is based on “**Passive**” components.

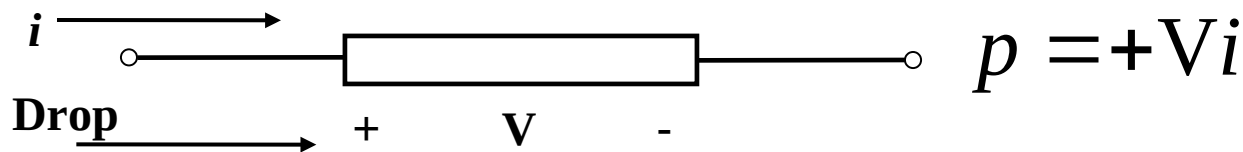
Implications

Variables are treated as positive when the polarities are consistent with the “**Passive Sign Convention**”

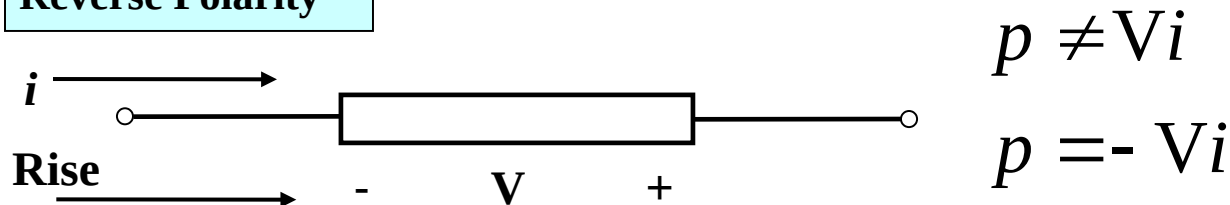
Example/Illustration

Taken as Fact: Power = Voltage x Current

Standard Polarity



Reverse Polarity

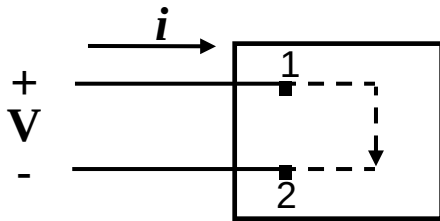


POSITIVE CURRENT WILL NOT FLOW IN THE REVERSE POLARITY FOR A PASSIVE COMPONENT

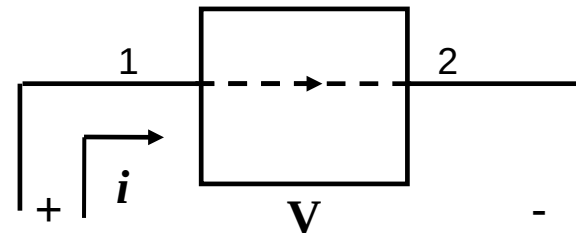
Ideal Circuit Elements

- 2 Terminal devices
- Described by V and/or i
- Basic Element

Two ways to represent this schematically



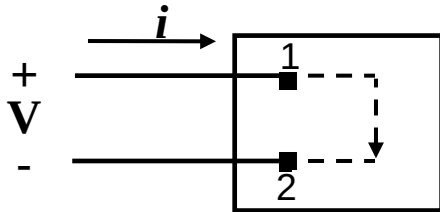
Nilsson and Reidel



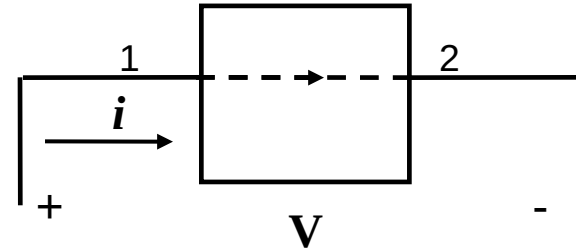
Harrell's

- Current “ i ” flows through the element.
- Voltage “ V ” is across the element.

Ideal Circuit Elements (Contd.)



Nilsson and Reidel



Harrell's

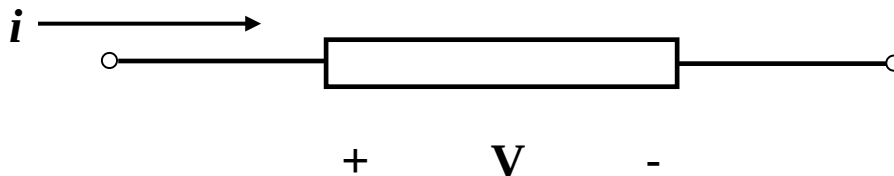
For positive V and i

➡ Voltage drop from 1 to 2 {+ to -}

➡ Voltage rise from 2 to 1

- i flows from 1 to 2: direction of + charge flow.
- “-” charge {electrons} flows opposite to i .

Passive Sign Convention



$$V = iR$$

Power and Energy

Power $\equiv$ Energy/Time

$$p = \frac{dW}{dt} \left\{ \text{Watt} = \frac{J}{s} \right\} = \frac{\text{Change in Energy}}{\text{Change in Time}}$$

$$p = \frac{dW}{dt} \equiv \left(\frac{dW}{dq} \right) \left(\frac{dq}{dt} \right) \left\{ \text{Use the fact that } \frac{dq}{dt} = i \right\}$$

$$V = \frac{dW}{dq}$$

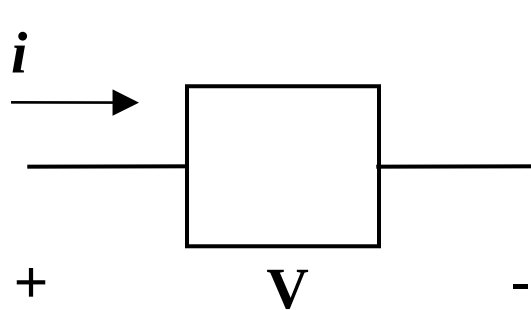
$$i = \frac{dq}{dt}$$

$$\therefore p = Vi$$

**Derived Power
Formula**

Power Delivery or Extraction

E
L
E
M
E
N
T



$$p = Vi$$

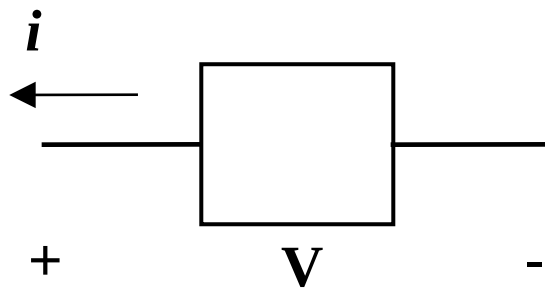
$\oplus \longrightarrow$ a drop $\Rightarrow$ lose energy

Power **delivered to OR** absorbed by element (Resistor)

$$p > 0$$

Power delivered to element

S
U
P
P
L
Y



$$p = -Vi$$

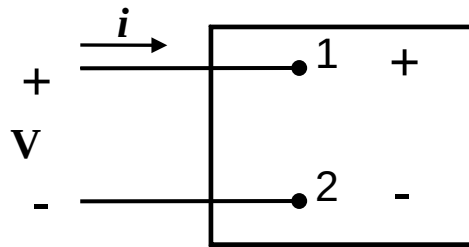
$\oplus \longrightarrow$ a rise $\Rightarrow$ gain energy

Power **extracted from OR** delivered by element (Power Supply)

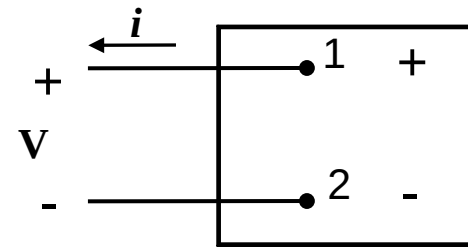
$$p < 0$$

Power extracted from element

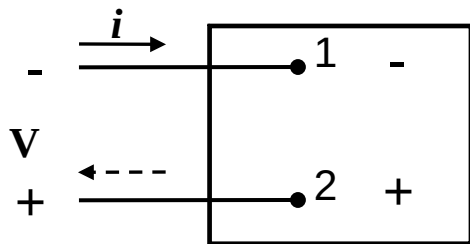
Find the Power Equation for the element in the box



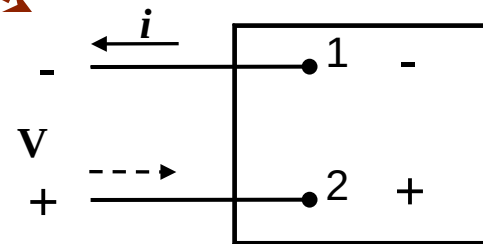
(a) $p = Vi$



(b) $p = -Vi$



(c) $p = -Vi$



(d) $p = Vi$

$$p > 0$$

Power Absorbed

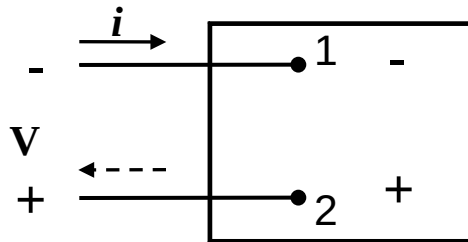
$$p < 0$$

Power Delivered

$$\sum p_{\text{Absorbed}} + \sum p_{\text{Delivered}} = 0 \quad \leftarrow \text{Conservation of Energy}$$

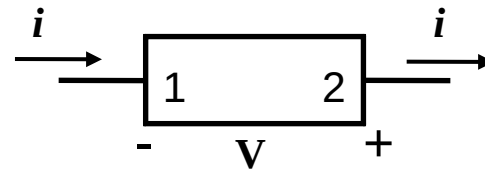
Example

From the previous figure part (c)



Let $V = 10 \text{ (V)}$
 $i = -2 \text{ (A)}$

1. Write equation:



$$p = -Vi$$

2. Plug in values:

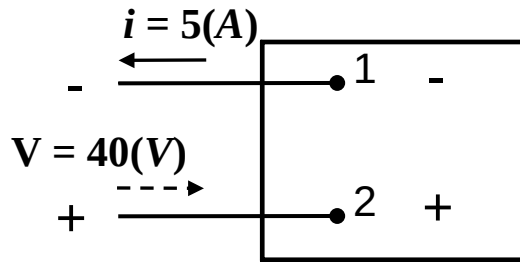
$$p = -(10)(-2) = +20 \text{ (W)}$$

$$p > 0$$

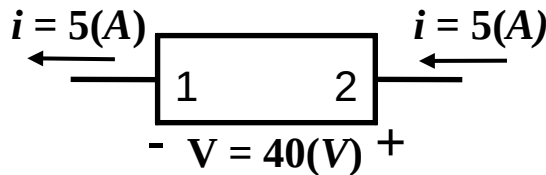


Power **absorbed** by element.

Example Problem



Redraw



Similar to Fig. (d)

Redraw as shown.

(a) Calculations:

$$\begin{aligned} p &= Vi \\ &= (40)(5) \\ &= 200 \end{aligned}$$

$p > 0$
Absorbed by box
Delivered to box

(b) Electrons leave terminal 2

(c) Electrons *lose energy*

Positive charges *lose energy*

Through
Voltage
drop

New Example:

$$V = -40 \text{ (V)}$$

$$i = 5 \text{ (A)}$$

then

Calculated or specified values

$$p = Vi = -200 \text{ (W)}$$

$p < 0$
Power Extracted **from** box